

Step 0:

Determine IP Address of server. Log in (however) and execute:

Bash

ip a

Look for lines like:

link/ether 52:e8:d4:c1:48:6b brd ff:ff:ff:ff:ff:ff

altname enp0s18

altname enx52e8d4c1486b

inet 147.79.74.113/24 brd 147.79.74.255 scope global eth0

Step 1: Upload Files

You need to customize the files before uploading. The lines you need to modify contain "?????" where the changes need to be made. These are Dockerfile, docker-compose.yml, DXVars.pm, and Listeners.pm.

In Dockerfile, you need to change the password and login name:

```
exec ttyd -p 8080 -W -c "sysop:???????" perl -I/spider/local -I/spider/perl /spider/perl/console.pl ??????\n\
```

In DXVars.pm:

```
$mycall = "????";  
$myname = "?????";  
$myalias = "???????"; # Example: W3LPL-9  
$mylatitude = "?????"; # Example: +39.3  
$mylongitude = "?????"; # Example: -77.0  
$mylocator = "?????"; # Example: FM19LG  
$myqth = "?????"; # Example: Glenwood, MD  
$myemail = "?????????";
```

In issue, change the login banner to whatever you want.

In motd, change the post-login prompt to whatever you want.

Use sftp as user root (sftp root@147.79.74.113), Upload all 6 files (setup_root.sh, setup_sysop.sh, Dockerfile, docker-compose.yml, DXVars.pm, and Listeners.pm) to /tmp on your server.

From Windows:

```
C:\Users\mbdev\OneDrive\Desktop\DXSpider\GoldFiles>sftp
root@147.79.74.113
The authenticity of host '147.79.74.113 (147.79.74.113)' can't be
established.
ED25519 key fingerprint is
SHA256:GW92tZVqZiLIw4IMuZF5syyqkxenG6Yr7pZjcXWdbKE.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])?
Warning: Permanently added '147.79.74.113' (ED25519) to the list of
known hosts.
root@147.79.74.113's password:
Connected to 147.79.74.113.
sftp> dir
sftp> pwd
Remote working directory: /root
sftp> cd /tmp
sftp> ll
Volume in drive C is Windows
Volume Serial Number is 4E4D-9B30
```

Directory of C:\Users\mbdev\OneDrive\Desktop\DXSpider\GoldFiles

```
2026-07-30 16:31 <DIR> .
2026-07-29 17:47 <DIR> ..
2026-07-29 23:45      304 docker-compose.yml
2026-07-29 23:45    1,855 Dockerfile
2026-07-29 23:45    1,758 DXVars.pm
2026-07-29 22:27   15,174 InstallationInstructions.docx
2026-07-30 16:31   16,154 InstallationInstructionsV2.docx
2026-07-29 23:46     496 Listeners.pm
2026-07-29 23:46    1,464 setup_root.sh
2026-07-29 23:46    1,832 setup_sysop.sh
                8 File(s)          39,037 bytes
                2 Dir(s)  1,226,850,512,896 bytes free
sftp> put Dockerfile
Uploading Dockerfile to /tmp/Dockerfile
Dockerfile
100% 1855    37.7KB/s   00:00
sftp> put DXVars.pm
Uploading DXVars.pm to /tmp/DXVars.pm
```

```
DXVars.pm
100% 1758    35.8KB/s   00:00
sftp> put docker-compose.yml
Uploading docker-compose.yml to /tmp/docker-compose.yml
docker-compose.yml
100% 304     6.9KB/s   00:00
sftp> mput setup*.sh
Uploading setup_root.sh to /tmp/setup_root.sh
setup_root.sh
100% 1464    31.8KB/s   00:00
Uploading setup_sysop.sh to /tmp/setup_sysop.sh
setup_sysop.sh
100% 1832    40.7KB/s   00:00
sftp> put Listeners.pm
Uploading Listeners.pm to /tmp/Listeners.pm
Listeners.pm
100% 496     10.1KB/s   00:00
sftp> ls
DXVars.pm
Dockerfile
Listeners.pm
docker-compose.yml
setup_root.sh
setup_sysop.sh
systemd-private-9e642035a3b447b3bae12f6d1d5af055-systemd-
logind.service-2cvx8M
sftp>
```

Step 2: Execute Root Setup

Log in as root (or use your default initial login):

On Windows:

```
C:\Users\mbdev\OneDrive\Desktop\DXSpider\GoldFiles>ssh
root@147.79.74.113
root@147.79.74.113's password:
Linux srv1849451 6.12.95+deb13-amd64 #1 SMP PREEMPT_DYNAMIC Debian
6.12.95-1 (2026-07-04) x86_64
```

The programs included with the Debian GNU/Linux system are free software;

[...]

```
root@srv1849451:~# cd /tmp
root@srv1849451:/tmp# chmod +x *.sh
root@srv1849451:/tmp# ./setup_root.sh
```

```
=== 1. Updating System & Installing Prerequisites ===
[...]
=== 3. Cleaning Up Default User & Creating Sysop User ===
Please set a password for the sysop user:
New password:
Retype new password:
passwd: password updated successfully
```

```
=====
==
Root setup complete!
Please log out completely and log back in as 'sysop'
then execute setup_sysop.sh
```

Step 3: Log Out

Log completely out of your SSH/terminal session.

Bash

exit

Step 4: Log In as sysop & Execute Sysop Setup

Log back in as the **sysop** user via SSH:

On Windows:

```
C:\Users\mbdev\OneDrive\Desktop\DXSpider\GoldFiles>ssh
sysop@147.79.74.113
sysop@147.79.74.113's password:
Linux srv1849451 6.12.95+deb13-amd64 #1 SMP PREEMPT_DYNAMIC Debian
6.12.95-1 (2026-07-04) x86_64
```

The programs included with the Debian GNU/Linux system are free software;

```
[...]
sysop@srv1849451:~$ cd /tmp
sysop@srv1849451:/tmp$ ls -al
total 28
drwxrwxrwt  7 root root  260 Jul 30 16:38 .
drwxr-xr-x 18 root root 4096 Jul 30 16:27 ..
-rw-r--r--  1 root root  304 Jul 30 16:32 docker-compose.yml
-rw-r--r--  1 root root 1855 Jul 30 16:32 Dockerfile
-rw-r--r--  1 root root 1758 Jul 30 16:32 DXVars.pm
drwxrwxrwt  2 root root   40 Jul 30 16:26 .font-unix
drwxrwxrwt  2 root root   40 Jul 30 16:26 .ICE-unix
```

```
-rw-r--r--  1 root root  496 Jul 30 16:32 Listeners.pm
-rwxr-xr-x  1 root root 1464 Jul 30 16:32 setup_root.sh
-rwxr-xr-x  1 root root 1832 Jul 30 16:32 setup_sysop.sh
drwx-----  3 root root   60 Jul 30 16:26 systemd-private-
9e642035a3b447b3bae12f6d1d5af055-systemd-logind.service-2cvx8M
drwxrwxrwt  2 root root   40 Jul 30 16:26 .X11-unix
drwxrwxrwt  2 root root   40 Jul 30 16:26 .XIM-unix
sysop@srv1849451:/tmp$
```

Bash

Then execute:

Bash

```
sysop@srv1849451:/tmp$ cd /tmp
sysop@srv1849451:/tmp$ echo $UID
1000
sysop@srv1849451:/tmp$ ./setup_sysop.sh
=== 1. Creating Production Directory Structure ===
```

```
=====
==
Please copy your 4 support files into /tmp:
- /tmp/DXVars.pm
- /tmp/Listeners.pm
- /tmp/Dockerfile
- /tmp/docker-compose.yml
=====
==
Press <Enter> once all 4 files are present in /tmp...
All 4 support files detected in /tmp!
=== 2. Copying Support Files into Staging Directory ===
=== 3. Building and Launching DXSpider Container ===
[...]
=====
==
DXSpider container deployment complete!
Web console running on port 8080.
Cluster Telnet running on port 7300.
```

Check that DXSPider is running:

```
sysop@srv1849451:/tmp$ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED
829344a77c59	dxspider-prod-dxspider	"/entrypoint.sh"	2 minutes ago

```

Up 2 minutes 0.0.0.0:7300->7300/tcp, [::]:7300->7300/tcp,
0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp dxspider
sysop@srv1849451:/tmp$ telnet localhost 7300
Trying ::1...
Connected to localhost.
Escape character is '^]'.
login: kd4d
Hello KD4D, this is W3LPL in Glenwood,MD
running DXSpider V1.57 build 686
Capabilities: ve7cc rbn
Nodes: 0/1 Users [Loc/Clr]: 1/1 Max: 0/1 - Uptime: 3m 8s
Please enter your name, set/name <your name>
Please enter your QTH, set/qth <your qth>
Please enter your location with set/location or set/qra
Please enter your Home Node, set/homenode <your home DX Cluster>
KD4D de W3LPL 30-Jul-2026 1649Z dxspider >
sh/user
Callsigns connected to W3LPL
KD4D
KD4D de W3LPL 30-Jul-2026 1649Z dxspider >
exit
Connection closed by foreign host.
sysop@srv1849451:/tmp$

```

From Windows check external access:

```

telnet 147.79.74.113 7300
login: kd4d
Hello KD4D, this is W3LPL in Glenwood,MD
running DXSpider V1.57 build 686
Capabilities: ve7cc rbn
Nodes: 0/1 Users [Loc/Clr]: 1/1 Max: 1/1 - Uptime: 5m 51s
Please enter your name, set/name <your name>
Please enter your QTH, set/qth <your qth>
Please enter your location with set/location or set/qra
Please enter your Home Node, set/homenode <your home DX Cluster>
KD4D de W3LPL 30-Jul-2026 1652Z dxspider >
bye

```

Connection to host lost.

C:\Users\mbdev\OneDrive\Desktop\DXSpider\GoldFiles>

And, from a web browser (<http://147.79.74.113:8080/>), using the credentials from DXVars.pm:

```
DXVars.pm:$myalias = "SYSOP";
```

And the password from Dockerfile:

```
Dockerfile: exec tttyd -p 8080 -W -c "sysop:usethispassword" perl -  
I/spider/local -I/spider/perl /spider/perl/console.pl W3LPL-9\n\
```

The command “sh/node” should work - this checks existence of the database.

Configure Skimmer Connections

In DXSpider Console (I used the web console:

<http://147.79.74.113:8080> with the DXVars.pm username in \$myalias and the password from Dockerfile in this line:

```
Dockerfile:  exec ttyd -p 8080 -W -c "sysop:usethispassword" perl
              -I/spider/local -I/spider/perl /spider/perl/console.pl W3LPL-9\n\
```

```
set/node sk0mmr
set/node sk1mmr
set/rbn sk0mmr sk1mmr
```

As user sysop:

Create the CW Connection Script in ~/DXSpider-prod/connect/sk0mmr

```
timeout 60
connect telnet telnet.reversebeacon.net 7000
'call:' 'W3LPL-RBNCW'
```

Create the FTx Connection Script in ~/DXSpider-prod/connect/sk1mmr

```
timeout 60
connect telnet telnet.reversebeacon.net 7001
'call:' 'W3LPL-RBNFT'
```

```
sysop@srv1849451:~/dxspider-prod/connect$ more sk*mmr
```

```
:::::::::::::
```

```
sk0mmr
```

```
:::::::::::::
```

```
timeout 60
```

```
connect telnet telnet.reversebeacon.net 7000
```

```
'call:' 'W3LPL-RBNCW'
```

```
:::::::::::::
```

```
sk1mmr
```

```
:::::::::::::
```

```
timeout 60
```

```
connect telnet telnet.reversebeacon.net 7001
```

```
'call:' 'W3LPL-RBNFT'
```

```
sysop@srv1849451:~/dxspider-prod/connect$ chmod +x sk*
```

```
sysop@srv1849451:~/dxspider-prod/connect$ ls -al
```

```
total 16
```

```
drwxrwxr-x 2 sysop sysop 4096 Jul 30 17:10 .
```

```
drwxrwxr-x 5 sysop sysop 4096 Jul 30 16:45 ..
```

```
-rwxrwxr-x 1 sysop sysop  80 Jul 30 17:10 sk0mmr
```

```
-rwxrwxr-x 1 sysop sysop  78 Jul 30 17:10 sk1mmr
```

```
sysop@srv1849451:~/dxspider-prod/connect$
```


To test before starting automatically:
From the DXSpider Console (I use the web console):

```
W3LPL-9 de W3LPL 30-Jul-2026 1713Z dxspider >
connect sk0mmr
connection to SK0MMR started
W3LPL-9 de W3LPL 30-Jul-2026 1713Z dxspider >
connect sk1mmr
connection to SK1MMR started
W3LPL-9 de W3LPL 30-Jul-2026 1714Z dxspider >
W3LPL-9 de W3LPL 30-Jul-2026 1717Z dxspider >
links
```

				Ave	Obs	Ping	Next
Filters							
Callsign	Type	Started	Uptime	RTT	Count	Int.	Ping
Iso?	In	Out	PC92?	Address			
	SK0MMR	RBN	30-Jul-2026 1717Z	2s		0	0
44.52.120.53							
	SK1MMR	RBN	30-Jul-2026 1714Z	2m 40s		0	0
44.10.10.51							

```
W3LPL-9 de W3LPL 30-Jul-2026 1717Z dxspider >
```

Copy issue and motd into ~/dxspider-prod/local_data. Make sure this
is done by user sysop.
